Lecture 4 v2:
OLS Implementation

25117 - Econometrics

Universitat Pompeu Fabra

October 9th, 2024

What we learned in the last lesson

- The population regression line, $\beta_0 + \beta_1 X$, is the mean of Y as a function of the value of X.
  - The slope, β_1, is the expected difference in Y between two observations with X values that differ by one unit.
  - The intercept, β_0, determines the level (or height) of the regression line.

- The population regression line can be estimated using sample observations (Y_i, X_i) with $i = 1, \ldots, n$ by ordinary least squares (OLS).
  - The OLS estimators of the regression intercept and slope are denoted $\hat{\beta}_0$ and $\hat{\beta}_1$.
  - The predicted value of Y given X is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$.

What we learned in the last lesson

- The R^2 and standard error of the regression (SER) are measures of how close the values of Y_i are to the estimated regression line.
  - The R^2 is between 0 and 1, indicating how much variance in our outcome is explained by the variance of our regressors.
  - The standard error of the regression estimates the standard deviation of the regression error.

- There are three key assumptions for estimating causal effects using the linear regression model:
  1. The regression errors, u_i, have a mean of 0, conditional on the regressors X_i;
  2. the sample observations are i.i.d. random draws from the population; and
  3. large outliers are unlikely

- If these assumptions hold, the OLS estimator $\hat{\beta}_1$ is
  1. an unbiased estimator of the causal effect β_1
  2. consistent
  3. normally distributed when the sample is large.

Estimation of the regression line

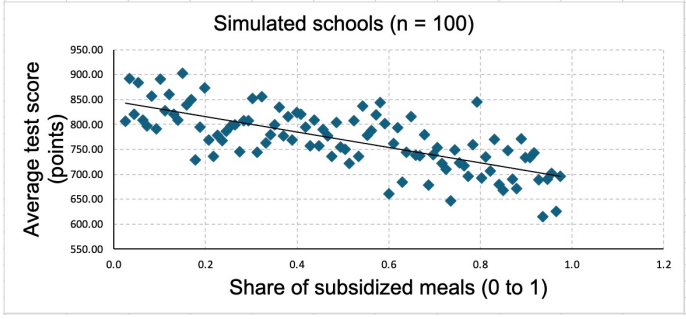
We want to learn about the slope of the population regression line. We have data from a sample, so there is sampling uncertainty. There are five steps toward this goal:

1. State the population of interest
2. Provide an estimator of this population object
3. Derive the sampling distribution of the estimator (this requires certain assumptions). In large samples, this sampling distribution will be normal by the CLT.
4. The square root of the estimated variance of the sampling distribution is the standard error (SE) of the estimator
5. Use the SE to construct t-statistics (for hypothesis tests) and confidence intervals.

Estimation of the regression line

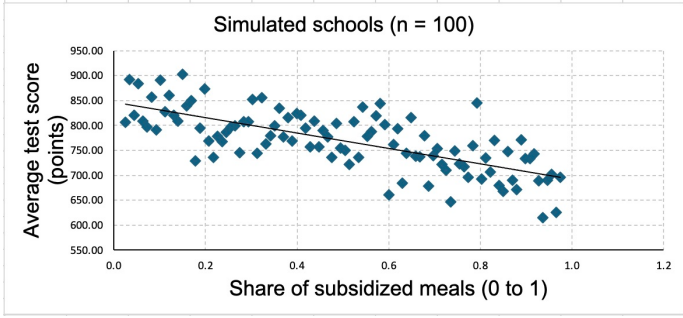
$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- β_1 is the slope of the population regression line
- $\hat{\beta}_1$ is the OLS estimator of β_1
- Under homoskedasticity and the usual regularity conditions, the large-sample distribution conditional on X is approximately $\mathcal{N}(\beta_1; \sigma_u^2 / TSS_X)$.



Hypothesis Testing

- Most of the time, we test for $H_0 : \beta_1 = 0$
- Null hypothesis and two-sided alternative:
  - $H_0 : \beta_1 = \beta_{1,0}$ vs. $H_1 : \beta_1 \neq \beta_{1,0}$
- Null hypothesis and one-sided alternative:
  - $H_0 : \beta_1 = \beta_{1,0}$ vs. $H_1 : \beta_1 < (or >)\beta_{1,0}$



Hypothesis Testing

- In general, the t-statistic is given by

$$t = \frac{\text{estimator} - \text{hypothetised value}}{\text{standard error of the estimator}}$$

- For μ_Y, the t-statistic is given by

$$t = \frac{\overline{Y} - \mu_{Y,0}}{s_Y/\sqrt{n}}$$

- For β_1, the t-statistic is given by

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)}$$

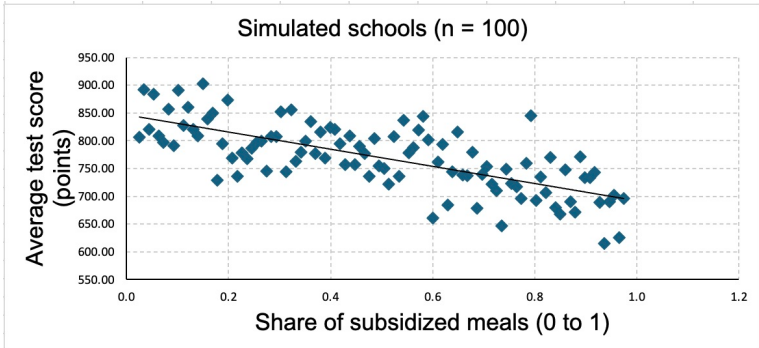
where $SE(\hat{\beta}_1) = \sqrt{\frac{SSR}{(n-2)TSS_X}} = \sqrt{\frac{\sum_{i=1}^n \hat{u}_i^2}{(n-2)\sum_{i=1}^n (X_i - \overline{X})^2}}$ when errors are homoskedastic (i.e., $var(u_i \mid X) = \sigma_u^2 \quad \forall i = 1, \ldots, n$)

- Using the large-sample normal approximation, reject $H_0 : \beta_1 = \beta_{1,0}$ at 5% if $|t| > 1.96$, or if the two-sided p-value is below 0.05. Excel's finite-sample test uses a t_{n-2} critical value instead.

Excel: A Dataset of 100 Simulated Schools

Open `Lecture4_Excel_100schools.xlsx`, sheet **Data**.

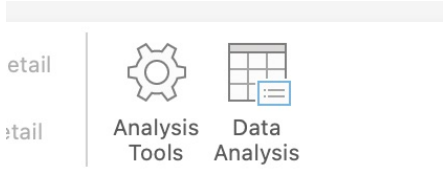
	A	B	C
1	Meal_share	Test_score	School_ID
2	0.0826	856.56	1
3	0.7447	749.42	2
4	0.1497	903.17	3
5	0.8311	770.40	4
6	0.8023	692.62	5
7	0.4760	736.09	6
8	0.4088	819.93	7
9	0.5432	836.38	8
10	0.3801	815.97	9
11	0.1402	808.97	10
12	0.4472	756.63	11
13	0.7927	844.84	12
14	0.8407	679.51	13
15	0.5816	844.02	14
16	0.1306	819.90	15
17	0.7063	753.72	16
18	0.6871	678.74	17
19	0.7159	722.32	18
20	0.4568	790.13	19
21	0.6775	779.54	20
22	0.5144	722.22	21
23	0.4952	754.44	22



- X_i: share of subsidized meals (0–1). Y_i: average test score, in points.
- These are **simulated teaching data**, with a negative association similar to the earlier example. They are not the original school records.

Excel Step 1: Enable the Analysis ToolPak

1. In desktop Excel, enable **Analysis ToolPak** if **Data Analysis** is missing:
   - **Mac:** Tools → Excel Add-ins → Analysis ToolPak → OK.
   - **Windows:** File → Options → Add-ins; Manage: Excel Add-ins → Go. Check Analysis ToolPak and click OK.
2. Open the **Data** tab and click **Data Analysis**.



The screenshots show Excel for Mac. The regression input ranges and calculations are the same on Windows.

Setup reference: Microsoft: Load the Analysis ToolPak in Excel.

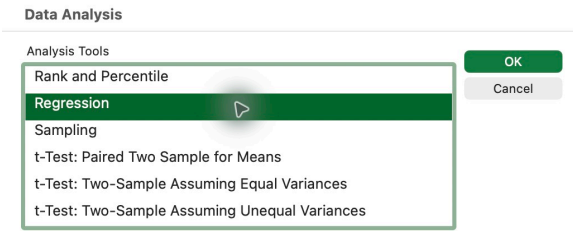
Excel Step 2: Choose Regression

1. In the list of analysis tools, select **Regression**.
2. Click **OK**.

We want Excel to estimate

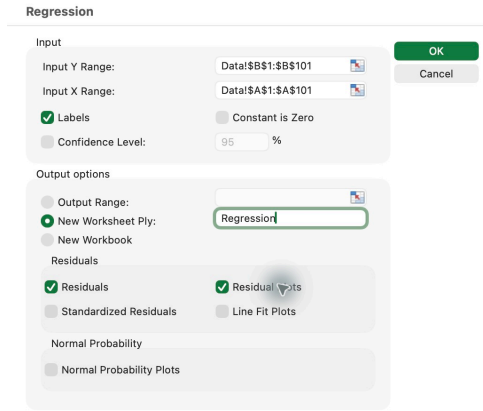
$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

by minimizing the sum of squared residuals.



Excel Step 3: Set the Input Ranges

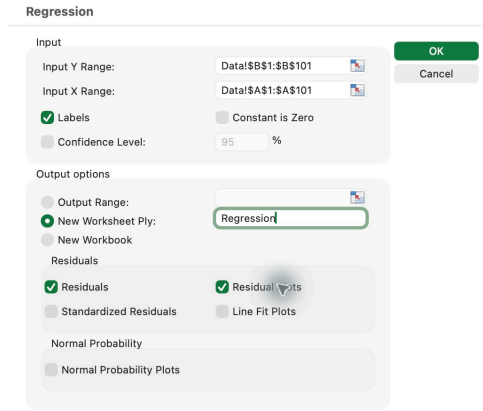
1. **Input Y Range:**
   `Data!$B$1:$B$101`
   The outcome is **Test_score**.
2. **Input X Range:**
   `Data!$A$1:$A$101`
   The explanatory variable is **Meal_share**.
3. Check **Labels**: row 1 contains names, so there are 100 observations.
4. Leave **Constant is Zero** unchecked: estimate an intercept.



Excel Step 4: Choose the Output

1. Choose **New Worksheet Ply** and name it **Regression**. If that sheet already exists, use a new name, such as **Regression2**.
2. Check **Residuals** and **Residual Plots**.
3. The output includes **95% confidence intervals** by default.
4. Click **OK**. Widen the output columns if numbers are cut off.

ToolPak output is a **snapshot**: rerun the regression after changing the data.



Excel Step 5: Read the Summary Output

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.713178796
R Square	0.508623996
Adjusted R Square	0.503609955
Standard Error	42.59996002
Observations	100

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	184088.797	184088.797	101.4399384	8.45674E-17
Residual	98	177846.1462	1814.756594		
Total	99	361934.9432			

- **Observations:** $n = 100$. Two coefficients are estimated, so residual degrees of freedom are $100 - 2 = 98$.
- **R Square:** 0.5086. **Standard Error:** 42.60, the SER.
- Under ANOVA, **Residual SS** is our *SSR*; **Total SS** is TSS_Y.

Yellow cells identify the results discussed in these slides.

Excel: Interpret the Estimated Coefficients

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	847.0719189	8.790754329	96.35941209	6.7995E-99	829.6269531	864.5168848
Meal_share	-154.8950979	15.37918018	-10.07173959	8.45674E-17	-185.4145804	-124.3756153

$$\widehat{Y}_i = 847.0719 - 154.8951 X_i.$$

- **Intercept:** the fitted score at $X = 0$ is 847.07. Since the simulated sample has $X > 0$, this is an extrapolation.
- **Slope:** increasing X by 0.10 (10 percentage points) is associated with a change of

$$-154.8951 \times 0.10 = -15.49 \text{ test-score points.}$$

- An association alone does not establish a causal effect of subsidized meals.

Excel: Test Whether Each Coefficient Is Zero

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	847.0719189	8.790754329	96.35941209	6.7995E-99	829.6269531	864.5168848
Meal_share	-154.8950979	15.37918018	-10.07173959	8.45674E-17	-185.4145804	-124.3756153

$$H_0 : \beta_j = 0, \qquad t_j = \frac{\widehat{\beta}_j - 0}{SE(\widehat{\beta}_j)}, \qquad p_j = P(|T_{98}| \geq |t_j|)\,.$$

Coefficient	Standard error	t statistic	Two-sided p-value
Intercept	8.7908	96.3594	6.80×10^{-99}
Meal_share	15.3792	-10.0717	8.46×10^{-17}

Both p-values are below 0.05: reject both zero-coefficient null hypotheses at 5%. These are very small probabilities, **not zero**.

ToolPak uses homoskedastic standard errors. Exact finite-sample t_{98} inference also assumes conditionally normal errors, with independent observations and conditional mean zero.

Excel: Construct 95% Confidence Intervals

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	847.0719189	8.790754329	96.35941209	6.7995E-99	829.6269531	864.5168848
Meal_share	-154.8950979	15.37918018	-10.07173959	8.45674E-17	-185.4145804	-124.3756153

$$\widehat{\beta}_j \;\pm\; t_{0.975,98}\, SE(\widehat{\beta}_j), \qquad t_{0.975,98} = 1.9845.$$

$$CI_{\beta_0} = 847.0719 \pm 1.9845(8.7908) = [829.6270,\ 864.5169],$$
$$CI_{\beta_1} = -154.8951 \pm 1.9845(15.3792) = [-185.4146,\ -124.3756].$$

- These are Excel's **Lower 95%** and **Upper 95%** columns (using unrounded values).
- Neither interval contains zero, consistent with the two-sided tests.
- In repeated sampling, 95% of intervals constructed this way cover the true coefficient, under the stated assumptions.

Excel: Standard Error, SER, and RMSE

First compute the fitted value and the residual:

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i, \qquad \widehat{u}_i = Y_i - \widehat{Y}_i, \qquad SSR = \sum_{i=1}^{100} \widehat{u}_i^2 = 177{,}846.1462.$$

Excel's Standard Error = SER

$$SER = \sqrt{\frac{SSR}{n-2}} = \sqrt{\frac{177{,}846.1462}{98}} = 42.6000.$$

The denominator adjusts for estimating two coefficients. This estimates the error standard deviation.

RMSE with denominator n

$$RMSE = \sqrt{\frac{SSR}{n}} = \sqrt{\frac{177{,}846.1462}{100}} = 42.1718.$$

This is the square root of the average squared prediction error within the sample.

Both are measured in **test-score points**. In this course, distinguish the denominators; the earlier Stata Root MSE label corresponds to the SER.

Excel: Compute and Interpret R^2

$$TSS_Y = \sum_i (Y_i - \overline{Y})^2 = 361{,}934.9432,$$

$$SSR = \sum_i (Y_i - \widehat{Y}_i)^2 = 177{,}846.1462,$$

$$ESS = \sum_i (\widehat{Y}_i - \overline{Y})^2 = 184{,}088.7970.$$

With an intercept, $TSS_Y = ESS + SSR$. Therefore,

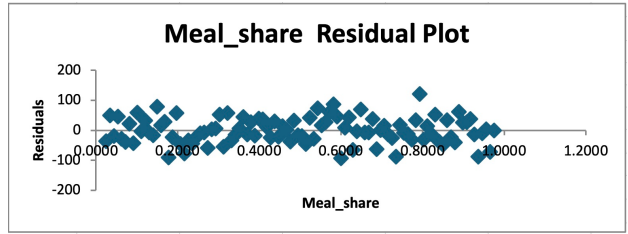
$$R^2 = 1 - \frac{SSR}{TSS_Y} = 1 - \frac{177{,}846.1462}{361{,}934.9432} = 0.5086.$$

- The fitted line explains **50.86% of the sample variation** in test scores around their mean.
- This measures in-sample fit, not the proportion of scores caused by meals.
- Adjusted $R^2 = 1 - (1 - R^2)\frac{n-1}{n-2} = 0.5036$ penalizes the use of a regressor.

Excel: Inspect Predicted Values and Residuals

RESIDUAL OUTPUT

Observation	Predicted Test_score	Residuals
1	834.2813013	22.27669867
2	731.7220042	17.69599576
3	823.8768427	79.29015729
4	718.344644	52.05135599
5	722.8037641	-30.18276409
6	773.3403034	-37.2543034
7	783.7449169	36.18008309
8	762.9356899	73.44031012
9	788.2038821	27.76511791
10	825.3630612	-16.38906117



For observation 1, $Y_1 = 856.5580$ and $\widehat{Y}_1 = 834.2813$, so

$$\widehat{u}_1 = Y_1 - \widehat{Y}_1 = 22.2767.$$

The model underpredicts this school's score by 22.28 points. Look for patterns or changes in residual spread; a plot alone cannot prove homoskedasticity.

By Hand, Step 1: Means and Deviations

Open **Paso_a_paso**. Each of the 100 observations has its own row (43–142).

$$\overline{X} = \frac{\sum_i X_i}{100} = 0.5000, \qquad \overline{Y} = \frac{\sum_i Y_i}{100} = 769.6244.$$

i	Xi	Yi	Xi - X-bar	Yi - Y-bar	(Xi-Xbar)^2	(Yi-Ybar)^2	(Xi-Xbar)(Yi-Ybar)
1	0.0826	856.5580	-0.4174	86.9336	0.1742	7,557.4560	-36.2882
2	0.7447	749.4180	0.2447	-20.2064	0.0599	408.2974	-4.9444
3	0.1497	903.1670	-0.3503	133.5426	0.1227	17,833.6340	-46.7737
4	0.8311	770.3960	0.3311	0.7716	0.1096	0.5954	0.2555
5	0.8023	692.6210	0.3023	-77.0034	0.0914	5,929.5190	-23.2760
6	0.4760	736.0860	-0.0240	-33.5384	0.0006	1,124.8223	0.8046
7	0.4088	819.9250	-0.0912	50.3006	0.0083	2,530.1534	-4.5855
8	0.5432	836.3760	0.0432	66.7516	0.0019	4,455.7801	2.8825

- Row 43: `D43=B43-$C$5` computes $X_1 - \overline{X}$; `E43=C43-$C$6` computes $Y_1 - \overline{Y}$.
- Then square each deviation and multiply the two deviations: `F43=D43^2`, `G43=E43^2`, `H43=D43*E43`.
- Copy down through row 142. The dollar signs keep the means fixed. Row 144 sums each column; both deviation sums are zero up to rounding.

By Hand, Step 2: From Sums to the OLS Line

Step	Concept	Value	Excel formula / explanation
1	Sample size n	100	=COUNT(Data!A2:A101)
2	Mean X (X-bar)	0.5000	=AVERAGE(Data!A2:A101)
3	Mean Y (Y-bar)	769.6244	=AVERAGE(Data!B2:B101)
4	Sum (Xi - X-bar)	0.0000	=SUM(D43:D142)
5	Sum (Yi - Y-bar)	0.0000	=SUM(E43:E142)
6	TSS_X	7.6728	=SUM(F43:F142)
7	Cross-product sum	-1,188.4740	=SUM(H43:H142)
8	Sample variance s_X^2	0.0775	=C9/(C4-1)
9	Sample covariance s_XY	-12.0048	=C10/(C4-1)
10	Slope b1-hat	-154.8951	=C10/C9
11	Intercept b0-hat	847.0719	=C6-C13*C5

Let $S_{XY} = \sum_i (X_i - \overline{X})(Y_i - \overline{Y})$ be the **cross-product sum**. Then $s_{XY} = S_{XY}/(n-1)$ and $s_X^2 = TSS_X/(n-1)$.

$$\widehat{\beta}_1 = \frac{S_{XY}}{TSS_X} = \frac{s_{XY}}{s_X^2} = \frac{-1188.4740}{7.6728} = -154.8951,$$

$$\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X} = 769.6244 - (-154.8951)(0.5000) = 847.0719.$$

Use `C13=C10/C9` and `C14=C6-C13*C5`. Keep full precision in the cells.

By Hand, Step 3: Predictions, Fit, and Error Size

For the first observation (row 43), enter:

Cell	Excel formula	Meaning
I43	`=$C$14+$C$13*B43`	$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i$
J43	`=C43-I43`	$\widehat{u}_i = Y_i - \widehat{Y}_i$
K43	`=J43^2`	$\widehat{u}_i^2$
L43	`=(I43-$C$6)^2`	$(\widehat{Y}_i - \overline{Y})^2$

Copy down through row 142. Sum column K for *SSR*, G for TSS_Y, and L for *ESS*.

12	SSR (sum residual^2)	177,846.1462	=SUM(K43:K142)
13	TSS_Y	361,934.9432	=SUM(G43:G142)
14	ESS (explained SS)	184,088.7970	=SUM(L43:L142)
15	R Square	0.5086	=1-C15/C16
16	Residual degrees of freedom	98	=C4-2
17	SER (Excel Standard Error)	42.6000	=SQRT(C15/C19)
18	RMSE (denominator n)	42.1718	=SQRT(C15/C4)

Then calculate $R^2 = 1 - SSR/TSS_Y$, $SER = \sqrt{SSR/98}$, and $RMSE = \sqrt{SSR/100}$.

By Hand, Step 4: Standard Errors and t Statistics

Under homoskedasticity, estimate the common error variance by SER^2:

$$SE(\widehat{\beta}_1) = \frac{SER}{\sqrt{TSS_X}} = \frac{42.6000}{\sqrt{7.6728}} = 15.3792,$$

$$SE(\widehat{\beta}_0) = SER\sqrt{\frac{1}{n} + \frac{\overline{X}^2}{TSS_X}} = 42.6000\sqrt{\frac{1}{100} + \frac{0.5000^2}{7.6728}} = 8.7908.$$

In **Paso_a_paso**, the formulas are:

- C22: `=C20/SQRT(C9)`; C23: `=C20*SQRT(1/C4+C5^2/C9)`.
- To test zero coefficients, divide each estimate by its own standard error:

$$t_1 = \frac{-154.8951}{15.3792} = -10.0717, \qquad t_0 = \frac{847.0719}{8.7908} = 96.3594.$$

- C24: `=C13/C22`; C25: `=C14/C23`.

By Hand, Step 5: p-Values and Confidence Intervals

A **p-value is a probability**: under H_0, how likely is a t statistic at least this extreme in either direction?

19	SE of slope b1-hat	15.3792	=C20/SQRT(C9)
20	SE of intercept b0-hat	8.7908	=C20*SQRT(1/C4+C5^2/C9)
21	t statistic: H0 b1 = 0	-10.0717	=C13/C22
22	t statistic: H0 b0 = 0	96.3594	=C14/C23
23	Two-sided p-value: b1	8.46E-17	=T.DIST.2T(ABS(C24),C19)
24	Two-sided p-value: b0	6.80E-99	=T.DIST.2T(ABS(C25),C19)
25	95% critical t value	1.9845	=T.INV.2T(0.05,C19)
26	Slope: lower 95%	-185.4146	=C13-C28*C22
27	Slope: upper 95%	-124.3756	=C13+C28*C22
28	Intercept: lower 95%	829.6270	=C14-C28*C23
29	Intercept: upper 95%	864.5169	=C14+C28*C23

Two-sided p-values ($df = 98$)

- Slope, C26:
  `=T.DIST.2T(ABS(C24),C19)`
- Intercept, C27:
  `=T.DIST.2T(ABS(C25),C19)`

95% confidence intervals

- Critical value, C28:
  `=T.INV.2T(0.05,C19)`
- Use $\widehat{\beta}_j \pm \texttt{C28} \times SE(\widehat{\beta}_j)$.

Yellow cells match ToolPak. Function names and separators may differ with Excel's language settings.

Homoskedasticity vs. Heteroskedasticity

- In regression analysis, we often make an important assumption about the variance of the error term, u, known as **homoskedasticity**.
- Homoskedasticity implies that the variance of the error term is constant for all observations.
- However, in most cases, this assumption may not hold, leading to **heteroskedasticity**, where the variance of u varies across observations.

Homoskedasticity vs. Heteroskedasticity

For homoskedastic data, we assume:

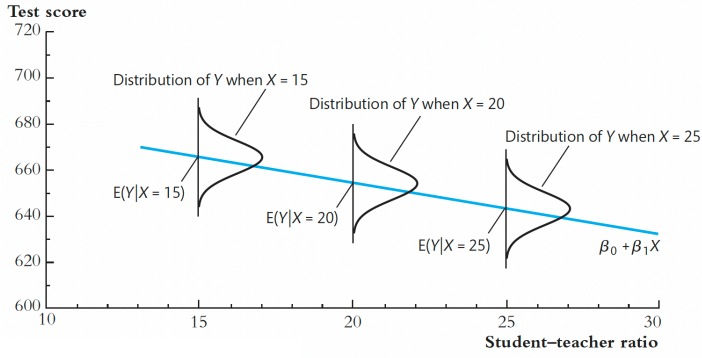
$$\text{var}(u_i \mid X) = \sigma^2 \quad \text{for all } i$$

For heteroskedastic data, the variance of u_i is not constant:

$$\text{var}(u_i \mid X) = \sigma_i^2 \quad \text{for each } i$$

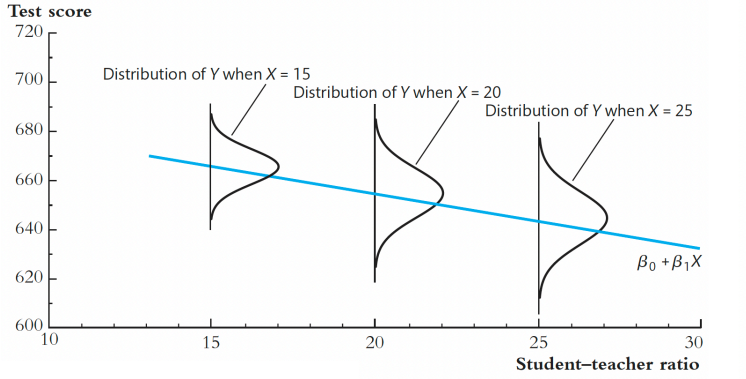
Graphical Illustration

- Understanding **homoskedasticity** and heteroskedasticity is crucial in regression analysis.
- **Homoskedasticity** assumes constant error variance, while heteroskedasticity implies varying error variance.
- To account for heteroskedasticity, robust standard errors can be used to obtain valid inferences.



Graphical Illustration

- Understanding homoskedasticity and **heteroskedasticity** is crucial in regression analysis.
- Homoskedasticity assumes constant error variance, while **heteroskedasticity** implies varying error variance.
- To account for **heteroskedasticity**, robust standard errors can be used to obtain valid inferences.



The Variance of the Slope with Heteroskedastic Errors

Write $TSS_X = \sum_i (X_i - \overline{X})^2$. Conditional on all the regressors, the weights are fixed:

$$\widehat{\beta}_1 - \beta_1 = \frac{\sum_i (X_i - \overline{X}) u_i}{TSS_X}.$$

If errors are uncorrelated across observations conditional on X,

$$\begin{aligned} \operatorname{var}(\widehat{\beta}_1 \mid X) &= \frac{1}{TSS_X^2} \operatorname{var}\left(\sum_i (X_i - \overline{X}) u_i \mid X \right) \\ &= \frac{\sum_i (X_i - \overline{X})^2 \operatorname{var}(u_i \mid X)}{TSS_X^2}. \end{aligned}$$

Under homoskedasticity, $\operatorname{var}(u_i \mid X) = \sigma_u^2$, so this reduces to σ_u^2 / TSS_X. With heteroskedasticity, the observation-specific variances remain inside the sum.

Estimate a Heteroskedasticity-Robust Standard Error

Use squared residuals to estimate each observation's contribution to the variance:

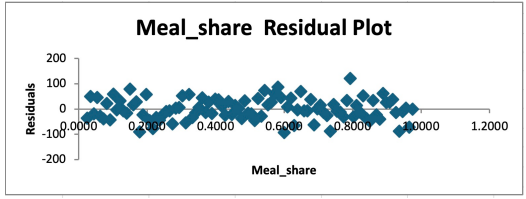
$$\widehat{\text{var}}_{\text{HC1}}(\widehat{\beta}_1) = \frac{n}{n-2} \frac{\sum_i (X_i - \overline{X})^2 \widehat{u}_i^2}{\left[\sum_i (X_i - \overline{X})^2\right]^2}.$$

- The numerator allows the size of the error to vary with X_i.
- The denominator is TSS_X^2: square the *sum of squared deviations*.
- $n/(n-2)$ is a degrees-of-freedom adjustment (HC1).
- Take the square root to obtain the robust standard error. Robust inference is justified by large-sample results under the usual regularity conditions.

The OLS coefficient estimates do not change; their estimated uncertainty does.

Excel in Practice: What the ToolPak Does Not Provide

- The standard errors in our **Regression** sheet use the homoskedasticity formula.
- The ToolPak Regression dialog has **no option for heteroskedasticity-robust standard errors**.
- If error variance changes with X, use a separate robust calculation before interpreting t tests and confidence intervals.



A residual plot helps diagnose changing spread. An absence of an obvious pattern does not establish constant variance.

The Theoretical Foundations of OLS

Let us consider a linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

Subject to the following assumptions:

1. Zero Conditional Mean: The errors u_i have a conditional mean of zero given the values of the independent variables, i.e., $\mathbb{E}(u_i|X_i) = 0$ for all i.
2. Random Draws: observations $(Y_i; X_i)$ are randomly drawn from the same population distribution, i.e., they are i.i.d. for all i.
3. Large outliers are rare: $\mathbb{E}(Y^4) < \infty$ and $\mathbb{E}(X^4) < \infty$
4. Homoskedasticity: The errors have constant variance, i.e., $\text{Var}(u_i) = \sigma_u^2$ for all i.
5. u is normally distributed: $u \sim \mathcal{N}(0; \sigma_u^2)$.

Gauss-Markov Theorem

Under assumptions 1-4, the OLS estimators are Best Linear Unbiased Estimators (BLUE).

The Gauss-Markov Theorem

- The OLS estimator $\hat{\beta}_1$ is a linear estimator, that is, it can be written as a linear function of $Y_1, ..., Y_n$:

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})u_i}{\sum_{i=1}^{n}(X_i - \overline{X})^2} \equiv w_i^{OLS} u_i$$

$$\text{where} \qquad w_i^{OLS} = \frac{\sum_{i=1}^{n}(X_i - \overline{X})}{\sum_{i=1}^{n}(X_i - \overline{X})^2}$$

- The Gauss-Markov theorem states that among all possible choices of $\{w_i\}$ the OLS weights w_i^{OLS} yield the smallest $var(\hat{\beta}_1)$.
- Under assumptions 1-5, the ordinary least squares (OLS) estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ have the minimum variance among all (linear and non-linear) unbiased estimators

The Gauss-Markov Theorem: Limits

The Gauss-Markov Theorem is a crucial justification for using OLS in many regression analyses, but it is very restrictive and, therefore, has serious practical limitations.

- The result is only for linear estimators –- only a small subset of estimators
- The condition of homoskedasticity often doesn't hold (homoskedasticity is a special case)
- When estimating the population mean, if there are significant outliers, the median is preferred over the mean because it's less sensitive and has lower variance. Similarly, in regression, OLS estimators can also be sensitive to outliers. In such cases, other estimators like the Least Absolute Deviations (LAD) estimator may be more efficient:

$$\min_{\hat{\beta}_0;\hat{\beta}_1} \sum_{i=1}^{n} \mid Y_i - \beta_0 - \beta_1 X_i \mid$$

Back to the original question

Suppose schools receive extra transfers from the central government so the share of subsidized school meals increases by 10%-pt. What is the effect of this policy intervention ("treatment") on test scores?

- This question requires an estimate of the causal effect on test scores of a change in the share of subsidized meals. Does our regression analysis using the California data set provide a compelling estimate of this causal effect?
- Not really – districts with a low share of subsidized meals tend to be ones with lots of other resources and higher income families, which provide kids with more learning opportunities outside school...this suggests that $corr(u_i; X_i) > 0$ (that is, $\mathbb{E}(u_i \mid X_i) \neq 0$).
- It seems that we have omitted some factors, or variables, from our analysis, and this has biased our results...

Material I

- **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapter 5